# Wellness Studio OS

Fitness Studio ERP & AI Growth Platform

## Master Product Requirements Document

| | |
|---|---|
| **Document** | Document 1 of 6 - Master PRD |
| **Version** | 1.0 (Release Candidate for Review) |
| **Date** | 29 September 2026 |
| **Status** | For CTO, Product Owner, Mentor, Investor and Engineering review |
| **Classification** | Confidential - Internal and Investor Use |

*This document is the official source of truth for product scope, requirements and priorities.*

# Table of Contents

**Document Control and Reading Guide . . . 5**

**1. Executive Summary . . . 6**

1.1 Product Overview . . . 6

1.2 Product Purpose . . . 6

1.3 Business Opportunity . . . 6

1.4 Product Positioning . . . 6

1.5 Product Value Proposition . . . 7

**2. Product Vision . . . 8**

2.1 Vision Statement . . . 8

2.2 Mission Statement . . . 8

2.4 Long-Term Product Vision . . . 8

2.3 Guiding Principles . . . 8

2.5 Future Evolution Strategy . . . 8

**3. Problem Statement . . . 9**

3.1 Existing Business Problems . . . 9

3.2 Studio Management Challenges . . . 9

3.3 Operational Challenges . . . 9

3.4 Customer Experience Challenges . . . 9

3.5 Growth Challenges . . . 9

**4. Business Objectives . . . 10**

4.1 Primary Objectives . . . 10

4.2 Secondary Objectives . . . 10

4.3 Growth Objectives . . . 10

4.4 Automation Objectives . . . 10

4.5 AI Objectives . . . 10

**5. Stakeholders . . . 11**

**6. User Personas . . . 13**

**7. User Roles . . . 14**

7.1 B2B Roles . . . 14

7.2 B2C Roles . . . 14

7.3 Platform Role . . . 15

7.4 Permission Matrix (Summary) . . . 15

7.5 Authentication Requirements . . . 15

**8. Product Scope . . . 16**

Part A - B2B Scope (Studio Management Portal) . . . 16

Part B - B2C Scope (Member Portal) . . . 17
Part C - AI Scope . . . 17
Part D - Automation Scope . . . 18
Part E - ERP Scope . . . 18
Explicitly Out of Scope for All Committed Phases . . . 18
**9. Functional Requirements . . . 19**
9.1 B2B - Studio Management Portal . . . 19
B2B-01 Dashboard . . . 19
B2B-02 Members . . . 19
B2B-03 Memberships . . . 20
B2B-04 Classes . . . 20
B2B-05 Bookings . . . 21
B2B-06 Attendance . . . 21
B2B-07 CRM . . . 21
B2B-08 Events . . . 22
B2B-09 Payments . . . 22
B2B-10 Reports . . . 23
B2B-11 Staff Management . . . 23
B2B-12 Settings (Configuration Center) . . . 23
B2B-13 Notifications . . . 24
B2B-14 Tenant Management . . . 24
B2B-15 AI Assistant . . . 25
9.2 B2C - Member Portal . . . 25
B2C-01 Dashboard . . . 25
B2C-02 Membership . . . 25
B2C-03 Bookings . . . 26
B2C-04 Attendance . . . 26
B2C-05 Events . . . 26
B2C-06 Payments . . . 26
B2C-07 Profile . . . 27
B2C-08 Notifications . . . 27
B2C-09 Recommendations . . . 27
**10. AI Features (MVP) . . . 28**
10.1 AI Assistant (Admin) . . . 28
10.2 AI Enquiry Assistant . . . 28
10.3 AI Booking Assistant . . . 28
10.4 AI Recommendation Assistant . . . 29
10.5 AI Feedback Assistant . . . 29

**11. Future AI Agents . . . 30**
11.1 CRM Agent (Phase 2) . . . 30
11.2 Operations Agent (Phase 2) . . . 30
11.3 Marketing Agent (Phase 3) . . . 30
11.4 Advertising Agent (Phase 3) . . . 30
11.5 Growth Optimization Agent (Phase 3) . . . 30
11.6 Studio Insights Agent (Phase 3) . . . 31
11.7 Session Summary Agent (Future) . . . 31
11.8 Agent Governance (applies to every agent) . . . 31
**12. Non-Functional Requirements . . . 32**
12.1 Security Requirements . . . 32
12.2 Performance Requirements . . . 32
12.3 Scalability Requirements . . . 32
12.4 Availability Requirements . . . 32
12.5 Reliability Requirements . . . 32
12.6 Audit Requirements . . . 33
12.7 Compliance Requirements . . . 33
12.8 Usability Requirements . . . 33
**13. Business Rules . . . 34**
13.1 Memberships . . . 34
13.2 Bookings . . . 34
13.3 Attendance . . . 34
13.4 Events . . . 34
13.5 Payments . . . 35
13.6 CRM . . . 35
13.7 Notifications . . . 35
13.8 AI Features . . . 36
13.9 Platform and General . . . 36
**14. Success Metrics and KPIs . . . 37**
14.1 Business KPIs . . . 37
14.2 Operational KPIs . . . 37
14.3 Membership KPIs . . . 37
14.4 Attendance KPIs . . . 37
14.5 Revenue KPIs . . . 37
14.6 AI Adoption KPIs . . . 38
14.7 Automation KPIs . . . 38
**15. Risks and Assumptions . . . 39**
15.1 Risks and Mitigation Plans . . . 39

15.2 Constraints . . . 39
15.3 Assumptions . . . 40
**16. Roadmap . . . 41**
16.1 Phase 1 - 5-Day Delivery (Current MVP) . . . 41
16.2 Phase 2 (indicative four to eight weeks after Phase 1) . . . 42
16.3 Phase 3 . . . 42
16.4 Future Product Evolution . . . 42
16.5 Agents by Phase . . . 43
**Appendix A - Conflict Resolution Register . . . 44**
**Appendix B - Glossary . . . 44**
**Appendix C - Open Decisions for Mentor and Business . . . 45**

# Document Control and Reading Guide

| Item | Detail |
|---|---|
| **Purpose** | Defines what Wellness Studio OS must do, for whom, in which order and to what standard. It is the official reference for scope, priorities, business rules and acceptance criteria. |
| **Audience** | CTO, Product Owner, Mentor, Investors, Solution Architects, Development Team, QA Team. |
| **Out of scope of this document** | Architecture design, database design, API specifications and execution plans. These are covered by separate documents in the documentation set. |
| **Priority notation** | **P1** = Phase 1 (5-Day Delivery, current MVP). **P2** = Phase 2. **P3** = Phase 3. **F** = Future product evolution. |
| **Requirement IDs** | Format: **B2B-XXX-nn** for studio management portal, **B2C-XXX-nn** for member portal, **AI-XXX-nn** for AI features, **BR-XXX-nn** for business rules, **NFR-XXX-nn** for non-functional requirements. |
| **Conflict handling** | Where inputs conflicted, the resolved product direction and justification are recorded in Appendix A. The body of this document reflects the resolved direction only. |
| **Change control** | Changes to frozen Phase 1 scope must be recorded, justified and approved by the mentor and the business stakeholder. |

# 1. Executive Summary

## 1.1 Product Overview

Wellness Studio OS is a multi-tenant Software-as-a-Service platform that lets dance, yoga, Zumba, meditation, fitness and other wellness businesses run their entire operation from a single system. It combines a **Studio Management Portal (B2B)** used by owners, administrators, front-desk staff and trainers with a **Member Portal (B2C)** used by members and prospective members. The platform manages leads, members, memberships, classes, bookings, attendance, events, payments, notifications and reporting, and layers an assistive AI capability on top of the same governed business data.

The platform is built as an API-first, agent-ready and automation-enabled product. An ERP system (ERPNext on the Frappe framework) is the system of record for business truth. Controlled APIs expose business capabilities to the user interfaces, the AI Assistant and automation workflows. Workflow automation (n8n) orchestrates repeatable processes and the WhatsApp Cloud API delivers member and lead communication. AI never reads or writes the database directly; it acts only through the same validated APIs as the human interface.

## 1.2 Product Purpose

The purpose of the product is to replace the fragmented spreadsheets, paper registers, ad-hoc messaging and disconnected point tools that studios use today with one configurable operating system. It is designed to improve operational efficiency, retention, utilisation and revenue, while giving studio owners trustworthy data and giving members a frictionless experience.

## 1.3 Business Opportunity

- **Underserved segment.** Independent studios and academies in India typically run on manual processes and messaging apps. Global fitness platforms are generally built for other markets, priced for larger operators and weak on local payment and messaging habits.
- **WhatsApp-centred communication.** Members and prospects in the target market communicate primarily through WhatsApp. Native, automated WhatsApp workflows are a strong differentiator.
- **Universal, configurable model.** One product serving dance, yoga, Zumba, meditation, gym and other wellness verticals through configuration widens the addressable market and lowers per-customer customisation cost.
- **Recurring revenue.** A tiered subscription (Basic, Premium, Ultra) plus setup fees, premium AI add-ons and a WhatsApp automation package creates predictable, expandable revenue per tenant.
- **AI as an upsell layer.** A governed Admin Assistant and later domain agents create differentiation and a natural premium tier without requiring risky, resource-intensive computer-vision investment.

## 1.4 Product Positioning

> **Positioning statement:** For wellness studio owners who are losing time and revenue to manual operations, Wellness Studio OS is a universal, AI-assisted studio operating system that unifies member lifecycle, scheduling, payments and WhatsApp automation in one configurable platform. Unlike single-vertical or overseas tools, it is India-first, WhatsApp-native and built so that AI acts safely through governed business capabilities.

| Positioning axis | Wellness Studio OS stance |
|---|---|
| **Category** | AI-powered studio operating system (operations, growth and intelligence), not just scheduling software. |
| **Market** | India-first (INR, English default), extensible to other currencies and languages. |
| **Differentiators** | Universal configurability; native WhatsApp automation; ERP-grade correctness; governed AI Assistant; multi-tenant SaaS with tenant branding. |
| **Deliberate non-goals** | Not a marketplace, not a native-mobile-first product, not a computer-vision coaching product in the current roadmap. |

## 1.5 Product Value Proposition

| For | Value delivered |
|---|---|
| **Studio owners** | One trusted view of members, revenue, occupancy and renewals; configuration without developers; reduced dependence on staff memory. |
| **Administrators and front desk** | Faster onboarding and booking, fewer manual reminders, fewer duplicate records, and an assistant that answers operational questions in seconds. |
| **Trainers** | Their sessions, participant lists and attendance in one place, without access to financial data. |
| **Members** | Live availability, instant confirmations, easy payment and renewal, and timely reminders on WhatsApp. |
| **Prospects** | Quick answers, simple registration for workshops and trials, and prompt follow-up. |
| **Investors** | Recurring SaaS revenue, defensible data and workflow moat, and a clear, de-risked path to differentiated AI capabilities. |

# 2. Product Vision

## 2.1 Vision Statement

> To be the operating system that every wellness studio runs on: universal in capability, configurable in behaviour and intelligent in assistance, with people always in control of consequential decisions.

## 2.2 Mission Statement

To remove the manual work of running a studio so that owners and trainers can focus on their members and their craft, by delivering reliable operations, automated communication and trustworthy insight in one platform.

## 2.4 Long-Term Product Vision

Over the long term, Wellness Studio OS evolves from an operational system of record into a growth and intelligence platform. Studios run their day-to-day operations on the core modules, rely on automation for repetitive communication, and use a family of governed domain agents that monitor operations, manage leads, draft campaigns and recommend optimisations. Multi-branch groups, franchises and white-labelled brands run on the same tenant model. The ERP remains the single source of business truth throughout.

## 2.3 Guiding Principles

| Principle | Meaning |
|---|---|
| **Reliability first** | Core operations are complete and dependable before intelligence is added. |
| **Configure, do not hard-code** | Plans, class types, capacities, roles, event categories, templates and timings are data managed by the tenant. |
| **One controlled backend** | The UI, the AI Assistant, automation and future agents use the same governed capabilities and the same validation. |
| **ERP is truth, AI assists** | Payments, memberships, bookings and attendance are deterministic business logic; AI can summarise, recommend and draft. |
| **Human in the loop** | Sensitive actions require confirmation; advertising spend requires explicit approval. |
| **Multi-tenant from day one** | Tenant isolation, branding and permissions are structural, not retrofitted. |
| **Everything traceable** | Every consequential human, automated and AI action records who, what, when and result. |

## 2.5 Future Evolution Strategy

| Horizon | Focus | Outcome |
|---|---|---|
| **H1 - Stabilise (Phase 1)** | Operational core, configuration, payments, WhatsApp automation (test mode), Assistant proof of value | A demonstrable, revenue-capable operational product on a multi-tenant foundation |
| **H2 - Optimise (Phase 2)** | Commercial completeness, production integrations, richer reporting, broader assistant and first domain agents | Production-grade product; measurable retention and utilisation gains |
| **H3 - Grow (Phase 3)** | Marketing, advertising and growth agents with human approval; predictive insight | AI-assisted acquisition and revenue optimisation |
| **H4 - Scale (Future)** | Multi-branch, franchise, white-label, hardware integration, native apps, marketplace | Platform for groups and franchises; broader ecosystem |
| **Research track** | Movement and posture intelligence for yoga and similar disciplines | Only after concept, IP and validation questions are resolved; not on the commercial roadmap |

# 3. Problem Statement

## 3.1 Existing Business Problems

Wellness studios operate recurring, capacity-limited services with memberships, trainer schedules, events and constant communication. Information about members, payments, attendance and messages is spread across calls, spreadsheets, paper registers, payment apps and chat threads. Staff spend their time reconciling information rather than serving members, and owners lack a trusted, current view of the business.

## 3.2 Studio Management Challenges

| Challenge | Consequence |
|---|---|
| **Fragmented member records with no single profile** | Duplicate members, missing history, wrong status at the front desk |
| **Membership rules kept in people's heads or spreadsheets** | Inconsistent eligibility decisions, disputes, revenue leakage |
| **Rules hard-wired into tools built for one studio** | Every change (new class type, new plan) needs a developer or a workaround |
| **Weak role separation** | Privacy and financial exposure; trainers seeing what they should not |
| **No reliable audit trail** | Attendance and payment corrections cannot be explained or trusted |

## 3.3 Operational Challenges

- Manual class coordination, overbooking and last-seat conflicts.
- Paper or chat-based attendance that cannot be analysed and is easy to dispute.
- Manual renewal follow-up, resulting in missed renewals and lapsed members.
- Manual payment reconciliation between gateway records, cash and membership status.
- Repeated staff effort answering the same enquiries and sending the same reminders.
- Dashboards that require many clicks and filters to answer simple questions such as which memberships expire this week.

## 3.4 Customer Experience Challenges

- Members cannot see live availability or their own membership status, and must call or message to book or renew.
- Confirmations and reminders are inconsistent or absent, increasing no-shows.
- Prospects who respond to an advertisement wait for manual replies and are often lost.
- Payments and receipts are hard to track from the member's side.

## 3.5 Growth Challenges

- Leads from advertisements and landing pages are not captured centrally, so acquisition cost and conversion cannot be measured.
- No structured follow-up sequence between enquiry, trial and membership purchase.
- Little visibility of retention risk, inactive members or under-utilised classes.
- No affordable way to introduce intelligent assistance while keeping business rules correct and auditable.
- Studios that want to add a new programme (for example Pilates) or a second location face rework in most existing tools.

# 4. Business Objectives

## 4.1 Primary Objectives

| ID | Objective | Measure of success |
|---|---|---|
| **BO-01** | Deliver a commercially deployable operational core (members, memberships, classes, bookings, attendance, events, payments, CRM) before investing in advanced AI. | Canonical end-to-end flow live for a pilot studio |
| **BO-02** | Establish a universal, configurable, multi-tenant product rather than a single-studio application. | Percentage of new-studio onboarding completed by configuration only |
| **BO-03** | Make the ERP the single source of business truth and expose all capabilities through controlled interfaces. | Zero direct data access by AI or automation; 100% of actions via governed interfaces |
| **BO-04** | Provide a reliable payment and membership lifecycle with clean reconciliation. | Payment reconciliation accuracy |

## 4.2 Secondary Objectives

- Reduce dependence on external intermediaries for communication by integrating WhatsApp directly.
- Keep technology choices (ERP, AI model provider) replaceable through clear boundaries.
- Create a reusable pattern (requirement, capability, interface, AI, automation, validation) for every future feature.
- Maintain a configuration layer so that business changes do not require deployments.

## 4.3 Growth Objectives

| ID | Objective | Measure |
|---|---|---|
| **GO-01** | Improve membership renewal rate through timely reminders and visible renewal pipelines. | Renewal rate; reminder-to-renewal conversion |
| **GO-02** | Improve class occupancy and reduce no-shows. | Occupancy rate; no-show rate |
| **GO-03** | Capture and convert external leads efficiently. | Lead-to-member conversion; source attribution coverage |
| **GO-04** | Grow SaaS revenue through tiers, setup fees and add-ons. | Tenants onboarded; recurring revenue; add-on attach rate |

## 4.4 Automation Objectives

- Automate predictable, repetitive communication: confirmations, reminders, renewal notices, event and workshop messages.
- Ensure automated workflows are traceable, retry-aware and failure-visible; failures create admin visibility rather than silent loss.
- Keep business logic in the platform and use the workflow engine only for orchestration.

## 4.5 AI Objectives

- Demonstrate an agent-ready architecture with a small number of high-value AI capabilities: an Admin Assistant that answers operational questions and executes one approved action, plus assistive enquiry, booking, recommendation and feedback features.
- Ensure AI operates only through governed interfaces, inherits the user's permissions, requires confirmation for transactions and is fully auditable.
- Avoid dependence on large historical datasets in the early product; rely on business rules, live ERP data, retrieval and existing foundation models.
- Preserve the option to adopt open-source or locally hosted models to manage cost and data-sovereignty concerns.

# 5. Stakeholders

The following stakeholders have a direct interest in, or are directly affected by, the product. Business, delivery and platform stakeholders are listed in Section 5.8.

### Studio Owner

| | |
|---|---|
| **Goals** | Understand business health; control configuration, roles and integrations; grow revenue and retention. |
| **Responsibilities** | Approves plans, pricing, roles and business rules; oversees reports and staff; owns commercial decisions. |
| **Pain points** | No trusted single view; dependence on staff memory; late or missing renewal insight; manual reconciliation. |
| **Success metrics** | Time to a complete business view; renewal rate; revenue per member; occupancy; onboarding of new programmes without developers. |

### Admin

| | |
|---|---|
| **Goals** | Run daily operations quickly and accurately; keep records correct; resolve exceptions. |
| **Responsibilities** | Manages members, memberships, schedules, bookings, events, routine payments, attendance corrections and staff scope. |
| **Pain points** | Too many screens; manual reminders; duplicate records; chasing renewals. |
| **Success metrics** | Time to onboard a member; time to answer an operational question; number of manual reminders eliminated; exception resolution time. |

### Front Desk Staff

| | |
|---|---|
| **Goals** | Serve walk-ins and enquiries quickly; book and check in members without errors. |
| **Responsibilities** | Registers members, records bookings on behalf of members, performs check-in, collects or records payments, captures enquiries. |
| **Pain points** | Slow lookups; duplicate members; unclear eligibility; queue pressure. |
| **Success metrics** | Registration time; check-in time; duplicate-record rate; booking error rate. |

### Trainer

| | |
|---|---|
| **Goals** | Deliver sessions efficiently with accurate participant information. |
| **Responsibilities** | Views assigned sessions and participants; marks attendance; performs approved trainer actions. |
| **Pain points** | No consolidated schedule; paper registers; unwanted exposure to financial data. |
| **Success metrics** | Attendance completeness for assigned sessions; time spent on admin per session. |

### Member

| | |
|---|---|
| **Goals** | Book, pay, attend and renew with minimal friction; know status at all times. |
| **Responsibilities** | Maintains own profile; books and cancels within rules; checks in; pays and renews; responds to reminders. |
| **Pain points** | Unknown availability; manual booking; missed renewals; unclear payment status. |
| **Success metrics** | Booking success rate; self-service share; reminder engagement; renewal rate. |

### Visitor (Prospect)

| | |
|---|---|
| **Goals** | Get quick, accurate answers and register for workshops, trials or membership. |
| **Responsibilities** | Submits enquiries and registrations; provides consent for follow-up. |
| **Pain points** | Slow replies; unclear timings and prices; complicated sign-up. |
| **Success metrics** | Time to first response; registration completion; enquiry-to-member conversion. |

### Management Team (Business and Delivery)

| | |
|---|---|
| **Goals** | Deliver a viable product, control scope and risk, and validate value with pilot studios. |
| **Responsibilities** | Product ownership, mentor and business approval, engineering, QA, AI and automation delivery, platform operations. |
| **Pain points** | Shifting requirements; ERP customisation effort; integration dependencies; credentials arriving late. |
| **Success metrics** | Phase gates met; defect escape rate; scope stability; pilot acceptance. |

# 6. User Personas

Personas are representative, fictional profiles derived from the target market and stakeholder needs. They guide prioritisation and design decisions.

### Neha Kapoor - Studio Owner (age 41)

| | |
|---|---|
| **Background** | Owns two yoga and Zumba studios in a metro city with about 400 active members and 12 staff. Previously ran the business on spreadsheets and a messaging app. Comfortable with smartphones, not technical. |
| **Goals** | See revenue, renewals and occupancy at a glance. Add a new class programme or plan without calling a developer. Control who can see what. Reduce dependence on one or two key staff. |
| **Pain points** | Cannot trust numbers reported by different staff. Discovers lapsed memberships too late. Worries about staff access to payments and member data. |
| **Product expectations** | A clean dashboard, simple configuration, strong role controls, monthly revenue and renewal insight, and an assistant she can simply ask questions. |

### Vikram Rao - Operations Manager (age 34)

| | |
|---|---|
| **Background** | Manages daily operations for a multi-programme academy. Handles schedules, exceptions, approvals, attendance corrections and renewals. Uses a laptop all day. |
| **Goals** | Fast member onboarding, no overbooking, accurate attendance, clear renewal pipeline, quick answers to management questions. |
| **Pain points** | Switches between many sheets and chats; manual reminders; time lost on corrections and reconciliation. |
| **Product expectations** | Efficient screens with filters, audited corrections, automated reminders with visible delivery status, and conversational queries such as which classes have low attendance. |

### Meera Nair - Trainer (age 29)

| | |
|---|---|
| **Background** | Leads Zumba and dance sessions across morning and evening slots. Uses her phone between sessions. |
| **Goals** | See today's sessions and participant list, mark attendance in seconds, know who is eligible. |
| **Pain points** | Paper attendance sheets; last-minute changes communicated through chat; no visibility of bookings. |
| **Product expectations** | A mobile-friendly trainer view limited to her sessions, with no access to finance. |

### Rahul Menon - Member (age 32)

| | |
|---|---|
| **Background** | Software professional who attends evening Zumba and weekend yoga. Prefers to do everything from his phone and expects instant confirmations. |
| **Goals** | Check availability live, book in seconds, pay online, get reminders, renew before expiry. |
| **Pain points** | Full classes with no waitlist; forgets renewal dates; unsure whether a payment succeeded. |
| **Product expectations** | A simple member portal, phone-OTP login, clear membership status, reliable WhatsApp confirmations and reminders, and sensible class suggestions. |

### Aisha Khan - Visitor (age 26)

| | |
|---|---|
| **Background** | Saw a yoga workshop advertisement on social media. Wants to compare timings and prices before committing. |
| **Goals** | Get answers immediately, register for a workshop or trial without creating a full account, and receive confirmation. |
| **Pain points** | Slow replies from studios; unclear price and timing information; too many form fields. |
| **Product expectations** | A short registration form, instant WhatsApp confirmation, a helpful enquiry assistant, and a friendly follow-up rather than silence. |

# 7. User Roles

Access is scoped by tenant, studio and branch, and enforced on both the interface and the underlying business capabilities. Roles are bundles of granular permissions. AI features inherit the permissions of the signed-in user and never exceed them. Each tenant's data, configuration and branding are isolated from every other tenant.

## 7.1 B2B Roles

### Owner

| | |
|---|---|
| **Responsibilities** | Business oversight; tenant configuration; roles and integrations; reporting; commercial decisions; staff oversight. |
| **Permissions** | Full access within the tenant, including configuration, roles, integrations, all reports, financial data and audit log. May use the AI Assistant for all permitted queries and actions. |
| **Access scope** | Entire tenant (all studios and branches owned). |

### Admin

| | |
|---|---|
| **Responsibilities** | Daily operations: members, memberships, classes, bookings, attendance, events, routine payments, CRM and exceptions. |
| **Permissions** | Create, read and update operational records; approve memberships where permitted; correct attendance with audit; limited configuration and staff scope; operational reports; AI Assistant. Cannot manage roles or integrations. |
| **Access scope** | Assigned studio(s) and branch(es). |

### Front Desk

| | |
|---|---|
| **Responsibilities** | Front-office work: enquiries, member onboarding, routine bookings on behalf of members, check-in and payment recording. |
| **Permissions** | Create, read and update members; book on behalf of members; check in; capture leads; record payments; limited reports. Cannot delete members or change configuration. AI Assistant read queries only, if enabled. |
| **Access scope** | Assigned studio or branch. |

### Trainer

| | |
|---|---|
| **Responsibilities** | Delivery of sessions: participant lists and attendance. |
| **Permissions** | Read assigned classes and participant lists; mark attendance for assigned sessions; approved trainer actions such as assisted booking only when explicitly granted. No financial data. |
| **Access scope** | Assigned classes and sessions only. |

## 7.2 B2C Roles

### Member

| | |
|---|---|
| **Responsibilities** | Self-service: manage own profile, view and renew membership, book and cancel, check in, register for events, pay, respond to notifications. |
| **Permissions** | Read and update permitted own-profile fields; read own memberships, bookings, attendance and payments; book eligible sessions; register for eligible events; make payments. No access to any other person's data. |
| **Access scope** | Own records only. |

### Visitor

| | |
|---|---|
| **Responsibilities** | Discover the studio and register interest. |
| **Permissions** | View public pages, classes and events; submit an enquiry; register for public workshops and events where allowed; use the public Enquiry Assistant. No login required. |
| **Access scope** | Public content and own submissions only. |

## 7.3 Platform Role

A **Platform Administrator** role exists on the operator side for tenant onboarding, tier entitlement and platform health. It has no routine access to tenant business data except through controlled, audited support procedures.

## 7.4 Permission Matrix (Summary)

| Capability | Owner | Admin | Front Desk | Trainer | Member | Visitor |
|---|---|---|---|---|---|---|
| **Tenant configuration** | Full | Limited | - | - | - | - |
| **Roles and integrations** | Full | - | - | - | - | - |
| **Staff management** | Full | Limited | - | - | - | - |
| **Members** | Full | Full | Create/Read/Update | Read (assigned) | Own | - |
| **Memberships** | Full | Full | Create/Renew* | - | Own (view/buy) | - |
| **Membership approval** | Yes | Yes* | - | - | - | - |
| **Classes and schedules** | Full | Full | View | Assigned | View/Book | Public view |
| **Bookings** | Full | Full | On behalf | Only if granted | Own | - |
| **Attendance** | Full | Mark/Correct | Check-in | Mark assigned | Own history | - |
| **Events** | Full | Full | Register others | Assigned | Register | Public events |
| **Payments** | Full | Operational | Record/Collect | - | Own | - |
| **CRM and leads** | Full | Full | Capture/Update | - | - | Submit enquiry |
| **Reports** | Full | Operational | Limited | Limited | - | - |
| **Audit log** | Read | Limited | - | - | - | - |
| **AI Assistant** | Full | Full | Read queries* | - | - | Enquiry only |

** Subject to explicit permission grant by the Owner. No role may bypass membership eligibility, payment or capacity rules.*

## 7.5 Authentication Requirements

- Staff (Owner, Admin, Front Desk, Trainer): email and password with mandatory password reset; multi-factor authentication for privileged staff before production release.
- Members: phone one-time passcode as the primary method, with optional Google sign-in; account linking must never create duplicate member identities.
- Visitors: no login for public content or permitted enquiry and registration flows.

# 8. Product Scope

Scope is separated by Phase 1 (5-Day Delivery, the current MVP), Phase 2, Phase 3 and Future. A capability marked P1 is committed for the 5-day delivery. The relative priority of P1 items, and what is cut first if time is short, is defined in Section 16.

## Part A - B2B Scope (Studio Management Portal)

| Capability | Phase 1 (MVP) | Phase 2 | Phase 3 / Future |
|---|---|---|---|
| **Dashboard** | Core KPI tiles, quick actions | Trend and pipeline views, drill-downs | Multi-branch control tower (F) |
| **Members** | Create, read, update, search, duplicate control | Bulk import, richer history | Engagement score (F) |
| **Memberships** | Studio plans, assignment, approval, expiry, manual renewal | Freeze, credits, upgrade and downgrade | Recurring debit opt-in (F) |
| **Classes and scheduling** | Configurable class types, sessions, trainer, capacity, recurring schedule | Blackout dates, substitution, rooms | Smart schedule optimiser (F) |
| **Bookings** | Eligibility, capacity, cancellation cutoff, waitlist, booking on behalf | Seat recovery, richer waitlist rules | Demand insight (F) |
| **Attendance** | QR, trainer marking, audited correction | No-show analytics | Hardware check-in (F) |
| **CRM** | Lead capture, status, conversion, source | Follow-up sequences, trial flow, attribution reports | Lead scoring, CRM Agent (P2/P3) |
| **Events** | Categories, capacity, free or paid, registration | Event waitlist, full reminder sequence | Feedback and check-in extensions (F) |
| **Payments** | Online payment (test mode), records, manual recording | Production gateway, refunds, invoices, reconciliation | Multiple providers (F) |
| **Reports** | Core operational reports | Drill-downs, exports, exception reports | Advanced analytics (F) |
| **Staff management** | Staff and roles, trainer assignment | MFA enforcement, granular grants | Commission and payroll (F) |
| **Settings** | Configuration Center basics, branding, templates | Extended configuration, integrations UI | White-label domains (F) |
| **Notifications** | Core WhatsApp workflows (test mode), delivery status | Retries console, email, preferences | Campaign channels (F) |
| **Tenant management** | Tenant, studio, branch structure, tier flags | Self-service onboarding and billing | Multi-branch and franchise (F) |
| **AI Assistant** | Two operational read queries and Create Member with confirmation | More queries and actions with approval queue | Cross-domain agent orchestration (P3) |

## Part B - B2C Scope (Member Portal)

| Capability | Phase 1 (MVP) | Phase 2 | Phase 3 / Future |
|---|---|---|---|
| **Dashboard** | Status, upcoming bookings, renewal prompt | Personalised highlights | Progress summaries (F) |
| **Membership** | Browse plans, buy, renew, view validity | Freeze request, upgrade | Family plans (F) |
| **Bookings** | Live availability, book, cancel, waitlist | Conversational booking | Personalised journey (F) |
| **Attendance** | QR check-in, history | Attendance summary view | Streaks (F) |
| **Events** | Browse, register, pay | Reminder set, waitlist | Feedback (F) |
| **Payments** | Checkout, history | Receipts and invoices | Saved methods (F) |
| **Profile** | Own details, phone OTP and Google login, linking | Communication preferences | Regional languages (F) |
| **Notifications** | WhatsApp confirmations and reminders | In-app centre, opt-in and out | Channel expansion (F) |
| **Recommendations** | Rules-constrained suggestions | AI-personalised suggestions | Goal-to-plan assistant (F) |
| **Public site** | Class and event discovery, enquiry and workshop registration | Trial booking | Referral programme (F) |

## Part C - AI Scope

| Capability | Phase 1 (MVP) | Phase 2 | Phase 3 / Future |
|---|---|---|---|
| **AI Assistant (admin)** | Read queries and Create Member with confirmation | Additional actions with approval queue, insights | Multi-agent orchestration |
| **AI Enquiry Assistant** | Approved-knowledge answers and lead capture | Multi-channel, richer handoff | Voice and regional language (F) |
| **AI Booking Assistant** | Guided suggestions handing off to standard booking | Conversational booking with confirmation | Proactive booking nudges (F) |
| **AI Recommendation Assistant** | Eligibility- and availability-constrained suggestions | Personalised by history and goals | Goal-to-plan (F) |
| **AI Feedback Assistant** | Sentiment, topic and summary | Trend dashboard | Automated follow-up workflows (F) |
| **Domain agents** | None (architecture is agent-ready) | CRM Agent, Operations Agent (read and draft) | Marketing, Advertising, Growth, Studio Insights, Session Summary |
| **Model strategy** | Commercial foundation model via API | Evaluate open-source or local model | Custom predictive models when data matures (F) |
| **Movement and posture intelligence** | Not in scope | Not in scope | Research track only (F) |

## Part D - Automation Scope

| Workflow | Phase 1 (MVP) | Phase 2 | Phase 3 / Future |
|---|---|---|---|
| **Registration and workshop confirmation** | Yes | Follow-up sequence | Adaptive sequences (F) |
| **Booking confirmation** | Yes | Waitlist promotion message | - |
| **Payment received and membership activated** | Yes | Receipts | - |
| **Class reminders** | Yes | Configurable rules UI | Smart timing (F) |
| **Renewal reminders (7, 3, 1 days and expiry day)** | Yes | Escalation and status check | - |
| **Event reminders** | Confirmation only | 24h, 6h, 1h reminders | - |
| **Inactive-member nudges** | No | 3 and 7 day nudges | Agent-recommended re-engagement (P3) |
| **Operational exception alerts** | Basic failure visibility | Admin task and alert workflow | - |
| **Delivery** | WhatsApp in test or mock mode | Production WhatsApp; email | Campaign channels (F) |

## Part E - ERP Scope

- **ERP platform:** ERPNext on the Frappe framework, hosted on a managed Frappe cloud environment. The upstream ERPNext code is never modified; all product-specific logic lives in a separate custom application.
- **System of record for:** members, memberships, class and session records, bookings, attendance, events and registrations, payment records, invoicing, and leads.
- **Access pattern:** the Next.js front ends never call the ERP directly; a single controlled API layer mediates every request and uses a restricted integration identity rather than an administrator account.
- **Replaceability:** ERP access is behind a defined boundary so the ERP or an individual capability can be replaced without changing the user interfaces or AI layer.
- **Application data store:** tenant, audit, AI and notification data that is not ERP-native is stored in the application database.
- **Not in Phase 1:** full accounting configuration, GST invoicing and advanced ERP reports (Phase 2).

## Explicitly Out of Scope for All Committed Phases

- Native mobile applications, marketplace and referral marketplace, community feed and gamification (Future).
- Camera-based posture and movement coaching, virtual coach and voice coaching (research track, not committed).
- Regional-language rollout and multi-currency (Future). Recurring auto-debit (evaluated in Phase 2).

# 9. Functional Requirements

Each requirement carries an identifier, description, business value, priority, user story and acceptance criteria. Priority: P1 = Phase 1 (5-Day Delivery), P2 = Phase 2, P3 = Phase 3, F = Future. Across all modules the following cross-cutting acceptance conditions apply: every create, update or delete is permission-checked, tenant-scoped and audit-logged; every screen has loading, empty, error and permission-denied states; and identifiers are generated by the system, never supplied by clients on creation.

## 9.1 B2B - Studio Management Portal

### B2B-01 Dashboard

***Purpose:*** *Give each role a fast, accurate view of operational health.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-DSH-01**<br>P1 | KPI tiles: active members, active memberships, today's attendance, monthly revenue, upcoming events, expiring memberships.<br>*Value: Fast, trusted overview; fewer navigation steps.* | As an Owner, I want to see key numbers on one screen so that I can judge business health at a glance. | • Given a signed-in Owner, all six tiles load and match the values from the underlying modules.<br>• Given no data, an empty state is shown, not an error.<br>• Values equal the answers given by the AI Assistant for the same period. |
| **B2B-DSH-02**<br>P1 | Quick actions: Add Member, Add Class, Add Event. Role-aware visibility of tiles and actions.<br>*Value: Shorter routine workflows.* | As a Front Desk user, I want quick actions on my home screen so that routine tasks take fewer clicks. | • Actions shown only if the user holds the permission.<br>• A Trainer sees only own sessions and no revenue data. |
| **B2B-DSH-03**<br>P2 | Occupancy, attendance and renewal pipeline (upcoming, overdue, completed) with drill-down; Send Campaign and Generate Report actions.<br>*Value: Retention and utilisation visibility.* | As an Admin, I want to see overdue renewals so that I can act before members lapse. | • Pipeline counts reconcile with membership records.<br>• Drill-down opens the filtered list. |

### B2B-02 Members

***Purpose:*** *Manage the full member lifecycle with one clean record per person.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-MEM-01**<br>P1 | Create, read, update, search and filter members with name, email, phone, date of birth, gender, join date, status (active, inactive, frozen), emergency contact, optional health notes and photo.<br>*Value: Single source of member truth.* | As a Front Desk user, I want to register a walk-in member in under a minute so that queues move. | • Valid request creates the member and returns a system-generated ID (format MEM-NNNNN).<br>• Missing name or email, invalid email, status or date format returns a clear field-level validation error.<br>• Clients cannot supply the ID. |
| **B2B-MEM-02**<br>P1 | Duplicate protection on email and phone; archive instead of destructive delete; role-limited deletion.<br>*Value: Prevents duplicate identities and data loss.* | As an Admin, I want a warning when a phone or email already exists so that we never create two records for one person. | • Duplicate email or phone is rejected with a conflict message.<br>• Front Desk cannot delete; Trainer is read-only; Member reads and updates only own record.<br>• All changes are audit-logged with before and after values. |
| **B2B-MEM-03**<br>P2 | Member history tabs (memberships, bookings, attendance, payments, communication) and bulk import.<br>*Value: Faster support, easier migration.* | As an Admin, I want to import existing members from a file so that onboarding a studio is fast. | • Import validates every row and reports rejects without partial silent failure.<br>• Imported members follow the same duplicate rules. |

## B2B-03 Memberships

***Purpose:*** *Define, sell, approve, track and renew studio member plans.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-MSP-01** P1 | Studio-defined plans (name, price, validity, allowed services, booking limits, active flag); assign a membership to a member; states Pending Approval, Active, Expired, Suspended, Cancelled. *Value: Configurable commercial model.* | As an Owner, I want to create a ten-class pass without a developer so that I can launch offers quickly. | • New plan takes effect without deployment. • End date must be after start date; price is zero or greater; member must exist. • Changing a plan does not alter existing memberships; the change is audited. |
| **B2B-MSP-02** P1 | Approval workflow with role sign-off; payment state and approval state kept separate; manual renewal; expiry tracking and expiring-soon list. *Value: Controlled activation; retention.* | As an Admin, I want to approve pending memberships so that only validated members get access. | • Confirmed payment alone does not activate a membership that requires approval. • Status changes to Expired automatically at end date. • Renewal creates new validity and a payment record. |
| **B2B-MSP-03** P2 | Freeze or pause, credit-based passes, upgrade and downgrade rules. *Value: Flexibility that reduces churn.* | As an Admin, I want to freeze a membership for a period so that travelling members are not lost. | • Freeze extends validity by the frozen duration. • Freeze and unfreeze are audited with reason. |

## B2B-04 Classes

***Purpose:*** *Define what the studio offers and schedule it.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-CLS-01** P1 | Configurable class and service types (Dance, Yoga, Zumba as seed; Gym, Meditation, Pilates and others added without code). *Value: Universal product; fast new programmes.* | As an Admin, I want to add a Pilates class type so that I can launch it without engineering. | • A new class type can be created, edited and archived without code or deployment. • Used class types are archived, not deleted. |
| **B2B-CLS-02** P1 | Sessions with date, time, trainer, capacity (default 30, overridable per class or session); recurring and one-off schedules; plan-to-class entitlement as configuration. *Value: Accurate scheduling and capacity control.* | As an Owner, I want different capacities for different rooms so that small classes are not overbooked. | • Capacity is read from configuration and never hard-coded. • Reducing capacity below confirmed bookings is blocked or requires explicit handling. • Trainer can be assigned per session. |
| **B2B-CLS-03** P2 | Holiday and blackout handling, trainer substitution, rooms and locations. *Value: Operational resilience.* | As an Admin, I want to substitute a trainer and notify booked members so that sessions still run. | • Substitution updates the roster and triggers a configurable notification. |

## B2B-05 Bookings

***Purpose:*** *Reserve seats with strict eligibility and capacity control.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-BKG-01** P1 | Member self-booking and booking on behalf by Admin or Front Desk; trainer-assisted booking only when permitted. Validates active membership, entitlement, booking limit, duplicates and capacity. *Value: No overbooking; less staff coordination.* | As a Front Desk user, I want to book on behalf of a walk-in member so that I can serve them immediately. | • A booking without an eligible membership is rejected with the reason.<br>• Two simultaneous requests for the last seat result in exactly one confirmation.<br>• A duplicate booking for the same member and session is rejected. |
| **B2B-BKG-02** P1 | Cancellation with configurable cutoff (default 4 hours) and waitlist with automatic promotion of the next eligible member; roster per session. *Value: Higher utilisation.* | As a Member, I want to join a waitlist for a full class so that I get a seat if one opens. | • Cancellation inside the cutoff is rejected and outside it releases the seat.<br>• Released seat promotes the next eligible waitlisted member automatically.<br>• Booking confirmation events are emitted only after the state is committed. |
| **B2B-BKG-03** P2 | Seat recovery rules and richer waitlist policies. *Value: Utilisation.* | As an Owner, I want late cancellations recovered so that empty seats are minimised. | • Recovery policy is configurable per class type. |

## B2B-06 Attendance

***Purpose:*** *Provide reliable, auditable per-session attendance.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-ATT-01** P1 | Attendance recorded per class or event session by QR check-in, trainer marking or front-desk check-in. *Value: Trusted engagement data.* | As a Trainer, I want to mark my class attendance on my phone so that records are accurate. | • A second check-in for the same member and session is rejected.<br>• A Trainer can mark only assigned sessions.<br>• Eligibility is checked; drop-in behaviour is configurable. |
| **B2B-ATT-02** P1 | Admin correction with immutable audit: actor, timestamp, previous value, new value, reason. *Value: Trust and dispute resolution.* | As an Admin, I want to correct a wrong record with a reason so that data stays trustworthy. | • Every correction stores all five audit fields and cannot be edited afterwards.<br>• Corrections appear in member attendance history. |
| **B2B-ATT-03** P2 | No-show analytics and inactivity patterns. *Value: Retention insight.* | As an Owner, I want to see inactivity patterns so that I can plan re-engagement. | • Inactivity list is available via reports and the AI Assistant. |

## B2B-07 CRM

***Purpose:*** *Capture prospects and convert them into members.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-CRM-01** P1 | Lead or Workshop Registrant created automatically from landing-page or advertisement forms, preserving source and campaign attribution; follow-up status. *Value: Measurable acquisition; no lost leads.* | As an Admin, I want every landing-page submission to appear as a lead so that nothing is missed. | • A submission creates a lead and triggers a confirmation without manual work.<br>• Leads never appear as members before conversion.<br>• Duplicate leads are flagged or merged. |
| **B2B-CRM-02** P1 | Explicit conversion of a lead to a member on a defined event (for example membership purchase) with history preserved. *Value: Clean prospect and member distinction.* | As an Admin, I want to convert a lead in one step so that no data is retyped. | • Conversion uses the standard member creation rules.<br>• Lead source and history are retained on the member. |
| **B2B-CRM-03** P2 | Optional trial-class flow, automated follow-up sequences, attribution reports. *Value: Higher conversion.* | As an Owner, I want to see which ad sources convert so that I can spend wisely. | • Conversion by source is reportable for any date range. |

## B2B-08 Events

***Purpose:*** *Run workshops, studio events and special sessions.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-EVT-01**<br>P1 | Event categories (Workshop, Internal Studio Event, Public Event, Special Session; configurable) with date, time, capacity, registration window, free or paid, and audience (public, members-only, prospects).<br>*Value: New revenue and lead source.* | As an Owner, I want to publish a paid workshop with 20 seats so that I earn beyond memberships. | • Registrations beyond capacity are blocked.<br>• Members-only events reject non-members.<br>• Registration triggers a confirmation. |
| **B2B-EVT-02**<br>P1 | Registration by members and external registrants; paid registration confirmed only after confirmed payment.<br>*Value: Revenue integrity.* | As a Visitor, I want to register for a public event without an account so that joining is easy. | • Pending payment does not consume a confirmed seat beyond the configured hold time.<br>• Confirmed payment converts registration to confirmed. |
| **B2B-EVT-03**<br>P2 | Event waitlist and full reminder sequence (24h, 6h, 1h); event attendance per session.<br>*Value: Higher attendance.* | As an Admin, I want automatic event reminders so that no-shows drop. | • Reminders fire at configured times without duplicates. |

## B2B-09 Payments

***Purpose:*** *Collect, record and reconcile payments.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-PAY-01**<br>P1 | Online payment through the payment gateway (test mode in Phase 1); payment record with internal ID, provider, provider reference, amount and currency (INR default), status (Created, Pending, Paid, Failed, Refunded), purpose and timestamps.<br>*Value: Revenue integrity.* | As a Member, I want to pay online and see my membership update so that I do not need to visit the studio. | • Only a verified server-side confirmation marks a payment Paid.<br>• Duplicate confirmations do not create duplicate payments.<br>• A delayed confirmation leaves the state Pending and never guesses success. |
| **B2B-PAY-02**<br>P1 | Manual or offline payment recording by Front Desk or Admin with audit.<br>*Value: Handles cash and assisted payments.* | As a Front Desk user, I want to record a cash payment so that the membership can be activated. | • Manual entries require an actor and reference and appear in reports. |
| **B2B-PAY-03**<br>P2 | Refunds, invoices with tax handling, reconciliation against gateway settlements, exception list.<br>*Value: Accounting readiness.* | As an Admin, I want mismatched payments highlighted so that I can resolve them. | • Reconciliation report lists unmatched and failed payments. |

## B2B-10 Reports

***Purpose:*** *Provide operational and financial insight.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-RPT-01**<br>P1 | Core reports: active members, revenue by plan and period, occupancy, attendance trend, renewal pipeline.<br>*Value: Evidence-based decisions.* | As an Owner, I want revenue by plan and month so that I can price better. | • Report totals reconcile with module data for the same period.<br>• Access respects role scope. |
| **B2B-RPT-02**<br>P2 | Drill-downs, payment exceptions, notification delivery and lead conversion reports; CSV and PDF export.<br>*Value: Depth and portability.* | As an Admin, I want to export a report so that I can share it with the owner. | • Exports respect the same filters and permissions as the screen. |
| **B2B-RPT-03**<br>P1 | All report data exposed through the same governed interface used by the AI Assistant.<br>*Value: Consistency across channels.* | As an Admin, I want the Assistant and dashboards to agree so that I trust both. | • Same metric and period give identical figures in dashboard and Assistant. |

## B2B-11 Staff Management

***Purpose:*** *Manage staff accounts, roles and trainer assignments.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-STF-01**<br>P1 | Create, deactivate and assign roles to staff with studio and branch scope; trainer profiles and class assignment.<br>*Value: Least-privilege security.* | As an Owner, I want to add a front-desk user with limited rights so that reception can work safely. | • Role changes take effect on the next request and are audited.<br>• Deactivated users cannot authenticate; historical records keep attribution. |
| **B2B-STF-02**<br>P1 | Granular permission grants such as trainer-assisted booking.<br>*Value: Flexible delegation.* | As an Owner, I want to grant a trainer booking rights for one studio so that they can assist members. | • Grant is scoped, revocable and audited. |
| **B2B-STF-03**<br>P2 | Multi-factor authentication enforcement for privileged staff; commission and payroll (F).<br>*Value: Security; enterprise readiness.* | As an Owner, I want MFA for all admins so that accounts are protected. | • MFA is required for Owner and Admin before production release. |

## B2B-12 Settings (Configuration Center)

***Purpose:*** *Externalise all changing business rules so no code change is needed.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-SET-01**<br>P1 | Studio profile and branding (name, logo, colours, contact details); member plans; class and service types; default capacity; cancellation window; event categories.<br>*Value: Universality; reduced change cost.* | As an Owner, I want my logo and colours in the member portal so that it feels like my brand. | • Branding changes need no deployment.<br>• All configuration changes are audited. |
| **B2B-SET-02**<br>P1 | Notification templates, trigger mapping and reminder timing; feature flags including AI features by tier.<br>*Value: Message and timing control.* | As an Owner, I want to change reminder timing and text so that messages fit my studio. | • Changes take effect without corrupting historical records.<br>• Onboarding a new wellness vertical requires configuration only. |
| **B2B-SET-03**<br>P2 | Integrations management screen and extended configuration.<br>*Value: Self-service administration.* | As an Owner, I want to manage integration credentials safely so that support requests decrease. | • Secrets are write-only in the UI and never displayed. |

## B2B-13 Notifications

***Purpose:*** *Deliver timely WhatsApp communication to members and leads.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-NTF-01** P1 | Configurable WhatsApp workflows: registration, payment received, membership activated, booking confirmation, class reminders, renewal reminders and expiry notice; delivery or failure status stored per message. *Value: Less manual work; fewer no-shows.* | As a Member, I want a booking confirmation and a reminder so that I do not miss class. | • Messages fire only after the business transaction commits.<br>• A failed message never reverses a valid transaction and is visible to admins.<br>• The same event never produces a duplicate message. |
| **B2B-NTF-02** P1 | Phase 1 runs in test or mock mode; production account credentials are supplied later without code change. *Value: Unblocks delivery.* | As an Admin, I want to see what would be sent so that I can validate before go-live. | • Mock mode stores a would-send record with recipient, template and time. |
| **B2B-NTF-03** P2 | Retry console, email channel, member communication preferences, inactive-member nudges. *Value: Reliability and reach.* | As an Admin, I want to retry failed messages so that important reminders are not lost. | • Retries are bounded and deduplicated; opted-out members receive no non-transactional messages. |

## B2B-14 Tenant Management

***Purpose:*** *Operate the platform as a multi-tenant SaaS.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-TEN-01** P1 | Tenant, studio and branch structure with strict isolation of data, configuration, branding and permissions; SaaS tier (Basic, Premium, Ultra) with feature entitlement flags. *Value: SaaS foundation and tier monetisation.* | As a Studio Owner, I want certainty that other studios cannot see my data so that I can trust the platform. | • Every business record is tenant-scoped; cross-tenant access attempts are denied and logged.<br>• Tier flags gate AI and automation features.<br>• A single pilot studio runs on the same model without special-casing. |
| **B2B-TEN-02** P1 | Operator-side tenant onboarding (seeded or assisted in Phase 1). *Value: Enables pilot.* | As a Platform Administrator, I want to create a tenant with an owner account so that a studio can start. | • Onboarding produces tenant, first studio, first branch and Owner user. |
| **B2B-TEN-03** P2 | Self-service onboarding, tenant billing for SaaS tiers. *Value: Scalable growth.* | As a new customer, I want to sign up and pay for a tier so that I can start quickly. | • Tier changes update entitlements immediately. |

## B2B-15 AI Assistant

***Purpose:*** *Let administrators query operations and perform approved actions in natural language.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2B-AIA-01** P1 | Natural-language operational queries answered from governed data, initially two curated queries: memberships expiring this week and classes with low attendance. *Value: Less dashboard navigation.* | As an Admin, I want to ask which memberships expire this week so that I get an answer without filters. | • Answer matches the dashboard for the same period and the user's permissions. • Unsupported questions receive a clear scope message, not a guess. |
| **B2B-AIA-02** P1 | Create Member action: intent recognition, field extraction, questions for missing mandatory fields, confirmation summary, execution through the same member creation capability as the UI, and reporting of the true result. *Value: Proves governed AI actions end to end.* | As an Admin, I want to say Add Rahul, Premium, starting Monday and be asked only for missing details so that onboarding is fast. | • No execution before explicit confirmation. • API validation errors and duplicates are reported truthfully; success is never claimed without success. • A user lacking the create permission cannot execute the action via the Assistant. • Relative dates are resolved and shown in the confirmation. |
| **B2B-AIA-03** P1 | Every Assistant call is audited with user, tool, arguments, result and timestamp; conversation context is temporary. *Value: Trust and traceability.* | As an Owner, I want every Assistant action logged so that I can audit it. | • Audit record exists for each read and write; context never becomes business data except through the governed capability. |

## 9.2 B2C - Member Portal

## B2C-01 Dashboard

***Purpose:*** *One place for a member's status and next steps.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2C-DSH-01** P1 | Membership status and validity, upcoming bookings, next class, recent attendance, renewal prompt and event highlights. *Value: Self-service; renewals.* | As a Member, I want to see my expiry and upcoming classes so that I can plan my week. | • Only the signed-in member's data is shown and status matches the ERP. • Membership expiring within seven days shows a renewal call to action. |

## B2C-02 Membership

***Purpose:*** *View, buy and renew plans.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2C-MSP-01** P1 | Browse active studio plans, purchase, renew and view validity, benefits and history. *Value: Online revenue capture.* | As a Member, I want to compare plans and buy one so that I can start attending. | • Only active plans are shown with tenant-configured price and validity. • Purchase creates a Pending membership until payment and any required approval succeed. |
| **B2C-MSP-02** P2 | Freeze request and upgrade request. *Value: Retention.* | As a Member, I want to request a freeze so that I do not lose my membership while travelling. | • Request enters the approval workflow and status is visible. |

## B2C-03 Bookings

***Purpose:*** *Find and book eligible classes.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2C-BKG-01** P1 | Live schedule with availability; book, cancel within window, join waitlist, view upcoming and past bookings. *Value: Occupancy and convenience.* | As a Member, I want to book the evening Zumba class and get instant confirmation so that my seat is secure. | • Ineligible classes show the reason (for example plan does not include Yoga). • Confirmation or rejection reflects the authoritative result, not cached state. • Cancellation shows and enforces the configured cutoff. |
| **B2C-BKG-02** P2 | Conversational booking with confirmation. *Value: Reduced friction.* | As a Member, I want to book by chatting so that it is faster on mobile. | • Booking is created only after member confirmation through the standard booking capability. |

## B2C-04 Attendance

***Purpose:*** *Check in and review history.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2C-ATT-01** P1 | QR check-in and attendance history reflecting admin corrections. *Value: Engagement transparency.* | As a Member, I want to scan a QR code to check in so that I do not queue. | • Check-in only for a valid session and eligibility; duplicates rejected with a clear message. |
| **B2C-ATT-02** P2 | Attendance summary and streak-style progress view. *Value: Motivation.* | As a Member, I want to see my monthly attendance so that I stay consistent. | • Summary is computed from authoritative attendance records. |

## B2C-05 Events

***Purpose:*** *Discover and register for events.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2C-EVT-01** P1 | Browse events by audience, register, pay if required and view own registrations. *Value: Event revenue and engagement.* | As a Member, I want to register for a weekend workshop and pay online so that my place is secure. | • Seat limit and registration window are enforced. • Members-only events are hidden from non-members. • Paid registration is confirmed only after verified payment. |
| **B2C-EVT-02** P2 | Reminder preference and event waitlist. *Value: Attendance.* | As a Member, I want to be reminded before an event so that I attend. | • Reminders follow configured timing. |

## B2C-06 Payments

***Purpose:*** *Pay and see payment history.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2C-PAY-01** P1 | Online checkout, payment history and clear status for own payments. *Value: Trust; fewer disputes.* | As a Member, I want to see whether my payment succeeded so that I am not charged twice. | • Only own payments are visible. • Failed or pending payments never display as paid. |
| **B2C-PAY-02** P2 | Receipts and invoices; retry of failed payments. *Value: Convenience.* | As a Member, I want a receipt so that I can keep records. | • Receipt matches the payment record. |

## B2C-07 Profile

***Purpose:*** *Self-service identity and preferences.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2C-PRF-01**<br>P1 | View and update permitted own details and emergency contact; phone one-time passcode login with optional Google sign-in and safe account linking.<br>*Value: Accurate data; low admin load.* | As a Member, I want to update my phone number safely so that reminders reach me. | • Members can edit only permitted fields.<br>• Login method changes re-verify identity and never create a duplicate member. |
| **B2C-PRF-02**<br>P2 | Communication preferences and consent management.<br>*Value: Compliance and trust.* | As a Member, I want to control which messages I receive so that I am not over-messaged. | • Preference changes apply to subsequent non-transactional messages. |

## B2C-08 Notifications

***Purpose:*** *Keep members informed on WhatsApp and in the app.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2C-NTF-01**<br>P1 | Receive WhatsApp confirmations and reminders for bookings, payments, membership and renewals.<br>*Value: Fewer no-shows; renewals.* | As a Member, I want reminders before class and expiry so that I do not miss either. | • Reminders fire at configured times and are not duplicated. |
| **B2C-NTF-02**<br>P2 | In-app notification centre and opt-in or opt-out controls.<br>*Value: Choice and reach.* | As a Member, I want an in-app history of notices so that I can review them. | • Opted-out members receive no non-transactional messages. |

## B2C-09 Recommendations

***Purpose:*** *Help members choose suitable classes.*

| ID / Priority | Description and Business Value | User Story | Acceptance Criteria |
|---|---|---|---|
| **B2C-REC-01**<br>P1 | Suggested classes based on stated goal, level, history and entitlement, constrained to eligible and available sessions.<br>*Value: Discovery and occupancy.* | As a beginner, I want suggestions for flexibility-focused classes so that I do not choose blindly. | • Only eligible and available classes are suggested, each with a short reason.<br>• Booking always goes through the standard booking flow. |
| **B2C-REC-02**<br>P2 | AI-personalised recommendations using attendance history.<br>*Value: Retention.* | As a Member, I want recommendations to improve as I attend more so that they stay relevant. | • Recommendation quality is measurable through acceptance rate. |

# 10. AI Features (MVP)

All MVP AI features share one design: a model-agnostic AI layer using an existing commercial foundation model through its API; the model is never trained from scratch and no large historical dataset is required. The AI layer uses only governed business capabilities as tools, retrieves only approved data, inherits the signed-in user's permissions and tenant scope, never accesses the database directly, and records an audit entry for every call. Deterministic business rules remain authoritative for payments, memberships, bookings and attendance.

> **MVP AI commitment for the 5-Day Delivery:** the Admin AI Assistant (two read queries and Create Member) is the flagship deliverable. The Enquiry, Booking, Recommendation and Feedback assistants are delivered in deliberately narrow form on top of interfaces that already exist that week. If the schedule slips, AI features are cut before any core operational module (see Section 16).

## 10.1 AI Assistant (Admin)

| | |
|---|---|
| **Purpose** | Let administrators ask operational questions and perform an approved action in natural language instead of navigating dashboards and filters. |
| **Business value** | Less administrative effort, faster decisions, and proof that AI can act safely through governed capabilities. |
| **User flow** | 1) Admin types a request. 2) Assistant identifies intent and the matching tool. 3) For queries: tool is called and a concise, human-readable answer is returned. 4) For Create Member: fields are extracted, missing mandatory fields are requested, a confirmation summary is shown, and on explicit confirmation the standard member creation capability is called. 5) The real success or failure result is reported. 6) The interaction is audited. |
| **Inputs** | Natural-language message; user identity, role and tenant; session context; tool definitions for members, memberships, attendance and analytics. |
| **Outputs** | Answer text derived from returned data; confirmation summary; action result with created record identifier or precise error; audit record. |
| **Success criteria** | At least 85% of curated queries answered correctly without escalation; at least 95% of confirmed actions correct on first confirmation; zero false-success reports; answers identical to dashboard figures; permission and tenant boundaries never exceeded. |

## 10.2 AI Enquiry Assistant

| | |
|---|---|
| **Purpose** | Provide first-line answers to prospects about classes, timings, plans and events and capture leads. |
| **Business value** | Faster response, lower staff workload and higher lead capture. |
| **User flow** | Visitor asks a question on the public site; the assistant answers from approved information; if the visitor wants to proceed it collects consent and contact details, creates a lead and offers registration; otherwise it escalates to a human. |
| **Inputs** | Visitor question; approved knowledge (public class, schedule, plan and event data; tenant-approved FAQs). |
| **Outputs** | Grounded answer; lead record through the CRM capability; handoff or escalation message. |
| **Success criteria** | Answers grounded in approved data with zero invented prices or promises; median response under 5 seconds; measurable lead capture and deflection rates. |

## 10.3 AI Booking Assistant

| | |
|---|---|
| **Purpose** | Help a member find a suitable session and reach the standard booking flow. |
| **Business value** | Higher booking conversion and lower support load. |
| **User flow** | Member states preference (class type, day, time); the assistant checks eligibility and availability and presents a shortlist with links into the standard booking flow. Conversational in-chat booking is Phase 2. |
| **Inputs** | Member intent; member eligibility; live availability from class and booking capabilities. |
| **Outputs** | Shortlist of eligible, available sessions with brief reasons. |
| **Success criteria** | 100% of suggested sessions are eligible and available at time of suggestion; no booking created without the member's confirmation through the standard flow. |

## 10.4 AI Recommendation Assistant

| | |
|---|---|
| **Purpose** | Suggest classes that fit the member's goal, level and history. |
| **Business value** | Discovery, occupancy and retention. |
| **User flow** | Member opens Recommendations or states a goal; the assistant applies eligibility and availability rules and returns ranked suggestions with a short explanation. |
| **Inputs** | Stated goal and level; past attendance; plan entitlement; live availability. |
| **Outputs** | Ranked, explainable suggestions. |
| **Success criteria** | Only eligible, available classes suggested; suggestion acceptance rate tracked; no medical or health claims. |

## 10.5 AI Feedback Assistant

| | |
|---|---|
| **Purpose** | Turn free-text post-class comments into structured insight for administrators. |
| **Business value** | Faster detection of recurring issues and positive themes. |
| **User flow** | Member submits a short comment after a session; the assistant classifies sentiment and topic and extracts the main issue or appreciation; administrators see individual results and a weekly theme summary. |
| **Inputs** | Free-text feedback with class or session reference; optional rating. |
| **Outputs** | Sentiment, topic, key point and a periodic theme summary. |
| **Success criteria** | Classification agrees with a human reviewer on at least 80% of a sampled set; raw comments always remain the record; no automatic action toward the member. |

# 11. Future AI Agents

Agents are modular and domain-specific. Each receives only the tools required by its domain, acts through governed capabilities, inherits user permissions, and requires human approval for sensitive actions. Agents collaborate through workflows when a process crosses domains; for example a booking cancellation can lead the Operations Agent to update capacity insight while the CRM Agent proposes a re-engagement follow-up. None of the agents below is part of Phase 1; the Phase 1 architecture only guarantees that they can be added without redesign.

## 11.1 CRM Agent (Phase 2)

| | |
|---|---|
| **Goal** | Manage the lead lifecycle and reduce lead leakage. |
| **Responsibilities** | Qualify and score leads; recommend follow-ups; draft follow-up messages for approval; track conversion. |
| **Inputs** | Lead records, source data, follow-up history, conversion events. |
| **Outputs** | Lead score, recommended next action, draft messages, conversion insight. |
| **Future business impact** | Higher lead-to-member conversion and less manual follow-up. |

## 11.2 Operations Agent (Phase 2)

| | |
|---|---|
| **Goal** | Monitor operational health continuously. |
| **Responsibilities** | Monitor capacity, attendance, bookings and memberships; raise alerts for low attendance or overbooked classes; propose schedule adjustments. |
| **Inputs** | Class, booking, attendance and membership data. |
| **Outputs** | Alerts, exception tasks, schedule suggestions. |
| **Future business impact** | Reduced manual monitoring and better utilisation. |

## 11.3 Marketing Agent (Phase 3)

| | |
|---|---|
| **Goal** | Improve engagement and retention through targeted communication. |
| **Responsibilities** | Draft WhatsApp and email campaigns for referral, re-engagement, event promotion and membership upsell; segment audiences; send only after approval. |
| **Inputs** | Member segments, renewal and activity data, event calendar. |
| **Outputs** | Draft campaigns, audience definitions, performance summaries. |
| **Future business impact** | Higher renewal rate, campaign engagement and event attendance. |

## 11.4 Advertising Agent (Phase 3)

| | |
|---|---|
| **Goal** | Acquire customers efficiently. |
| **Responsibilities** | Draft advertisement copy for social and search platforms; suggest audiences and landing pages; analyse campaign performance. Human approval is mandatory before publication or any spend. |
| **Inputs** | Lead source and conversion data, offers, event calendar. |
| **Outputs** | Ad drafts, audience suggestions, optimisation recommendations. |
| **Future business impact** | Lower customer acquisition cost and better lead quality. |

## 11.5 Growth Optimization Agent (Phase 3)

| | |
|---|---|
| **Goal** | Maximise revenue and utilisation. |
| **Responsibilities** | Analyse revenue, occupancy and conversion; recommend pricing and schedule changes; advisory only. |
| **Inputs** | Revenue, occupancy, conversion and plan data. |
| **Outputs** | Ranked, explained recommendations. |
| **Future business impact** | Revenue growth, occupancy improvement and membership upgrades. |

## 11.6 Studio Insights Agent (Phase 3)

| | |
|---|---|
| **Goal** | Give owners proactive business intelligence and forecasts. |
| **Responsibilities** | Summarise trends, detect anomalies, explain drivers of change, and (once data matures) produce churn and demand forecasts. |
| **Inputs** | Cross-module operational and financial data. |
| **Outputs** | Periodic insight briefings, anomaly alerts, forecasts. |
| **Future business impact** | Better strategic decisions; early warning on retention and demand. |

## 11.7 Session Summary Agent (Future)

| | |
|---|---|
| **Goal** | Turn attendance and approved notes into readable summaries. |
| **Responsibilities** | Generate member progress recaps and trainer or admin session summaries. |
| **Inputs** | Attendance, session and approved trainer notes. |
| **Outputs** | Member-facing and staff-facing summaries. |
| **Future business impact** | Higher engagement and retention; less trainer reporting effort. |

## 11.8 Agent Governance (applies to every agent)

- Agents never access the database directly; all reads and writes use governed capabilities.
- Each agent receives only the capabilities required by its domain; capabilities inherit the invoking user's permissions.
- Sensitive actions produce a draft and require human approval before execution.
- Every agent action is logged with initiator, capability, arguments, decision, approval, result and timestamp.
- Advertising publication and budget spend always require explicit human approval.

# 12. Non-Functional Requirements

## 12.1 Security Requirements

| ID | Requirement |
|---|---|
| **NFR-SEC-01** | All traffic uses HTTPS; all interfaces require authentication; authorisation (role, permission, tenant, branch and resource ownership) is enforced on the server for every request. |
| **NFR-SEC-02** | Staff use email and password with mandatory reset; multi-factor authentication is required for privileged staff before production. Members use phone one-time passcode with optional Google sign-in. |
| **NFR-SEC-03** | Secrets (database, token, payment, messaging and AI keys) are held in a secret store and never in source control or client code. |
| **NFR-SEC-04** | All inputs are validated; rate limiting, cross-origin controls, security headers and request-size limits are applied. |
| **NFR-SEC-05** | Sensitive data is encrypted in transit and at rest; personal data is minimised in AI prompts and logs. |
| **NFR-SEC-06** | The ERP is accessed through a restricted integration identity, never an administrator account; the browser never calls the ERP directly. |
| **NFR-SEC-07** | Payment status changes only through server-verified gateway confirmations; the client is never trusted. |

## 12.2 Performance Requirements

| ID | Requirement |
|---|---|
| **NFR-PRF-01** | Booking, check-in and payment initiation: 95th percentile response under 1 second under normal load. |
| **NFR-PRF-02** | Dashboards and standard reports load in under 3 seconds for typical single-studio data volumes. |
| **NFR-PRF-03** | Last-seat contention never produces overbooking; exactly one concurrent request succeeds. |
| **NFR-PRF-04** | AI Assistant read queries return within 8 seconds at the 95th percentile, subject to model provider latency. |

## 12.3 Scalability Requirements

| ID | Requirement |
|---|---|
| **NFR-SCL-01** | Every business record is scoped by tenant, studio and branch from day one so that multi-studio and multi-branch operation needs no data redesign. |
| **NFR-SCL-02** | The API layer is stateless and horizontally scalable; caching and queuing can be introduced without changing modules. |
| **NFR-SCL-03** | Components (front ends, API layer, ERP, AI layer, workflow engine) can be separated and scaled independently as load grows. |

## 12.4 Availability Requirements

| ID | Requirement |
|---|---|
| **NFR-AVL-01** | Target availability of 99.5% during beta and 99.9% in production, excluding planned maintenance. |
| **NFR-AVL-02** | Outage of the workflow engine or messaging provider must not block business transactions; events are queued and retried. |
| **NFR-AVL-03** | If the ERP is unavailable the product shows a clear temporary failure and never fabricates success or stale confirmations. |
| **NFR-AVL-04** | Daily backups with a tested restore procedure; rollback plan for every release. |

## 12.5 Reliability Requirements

| ID | Requirement |
|---|---|
| **NFR-REL-01** | Requests and callbacks are idempotent; duplicate payment or messaging callbacks create no duplicate transactions. |
| **NFR-REL-02** | Business events are emitted only after the transaction is committed; messaging failures never reverse valid transactions. |
| **NFR-REL-03** | Automated workflows retry with back-off, deduplicate and surface persistent failures to administrators. |

## 12.6 Audit Requirements

| ID | Requirement |
|---|---|
| **NFR-AUD-01** | An immutable audit record is written for approvals, payment changes, attendance corrections, role and permission changes, configuration changes and deletions. |
| **NFR-AUD-02** | Each record captures tenant, actor, actor type (human, automation, AI), action, resource, before and after values, timestamp and request context. |
| **NFR-AUD-03** | AI audit records additionally capture the request, capability invoked, arguments, decision, approval and result. |
| **NFR-AUD-04** | Correlation identifiers link an action across interface, API, ERP and workflow engine. |

## 12.7 Compliance Requirements

| ID | Requirement |
|---|---|
| **NFR-CMP-01** | Personal data handling aligns with applicable Indian data protection law: purpose limitation, consent, retention and deletion handling. |
| **NFR-CMP-02** | Messaging complies with the messaging provider's policies: approved templates, opt-in for marketing-type messages and honour of opt-outs. |
| **NFR-CMP-03** | Payments follow gateway and regulatory requirements; recurring debit, if introduced, is opt-in and compliant with authorisation rules. |
| **NFR-CMP-04** | Tax and invoicing (for example GST) are handled through the ERP's India compliance capabilities in Phase 2. |

## 12.8 Usability Requirements

| ID | Requirement |
|---|---|
| **NFR-USB-01** | Mobile-first, responsive interfaces for both portals; trainer and member flows work well on phones. |
| **NFR-USB-02** | Every screen provides loading, empty, error and permission-denied states with actionable messages. |
| **NFR-USB-03** | English is the default language; the product is internationalisation-ready for later regional languages. |
| **NFR-USB-04** | Administrative tasks give the same outcome whether performed through the interface or the AI Assistant. |

# 13. Business Rules

## 13.1 Memberships

| ID | Rule |
|---|---|
| **BR-MSP-01** | Studio member plans are defined by each tenant (name, price, validity, allowed services, booking limits, active flag). Platform subscription tiers (Basic, Premium, Ultra) are separate and never mixed with member plans. |
| **BR-MSP-02** | Membership states are Pending Approval, Active, Expired, Suspended and Cancelled. Payment status and approval status are independent. |
| **BR-MSP-03** | End date must be after start date; price must be zero or greater; the member must exist. |
| **BR-MSP-04** | Phase 1 uses manual renewal and payment. Reminders are sent 7, 3 and 1 days before expiry and on the expiry day. Recurring debit is a later, opt-in capability. |
| **BR-MSP-05** | A member with an expired, suspended or non-entitled membership cannot book restricted classes. |
| **BR-MSP-06** | Edits to an active plan are audited and do not silently change existing memberships. |

## 13.2 Bookings

| ID | Rule |
|---|---|
| **BR-BKG-01** | The backend and ERP perform the final validation of eligibility, booking limit, duplicates and capacity at commit time; the interface cannot bypass it. |
| **BR-BKG-02** | Default capacity is 30, stored as configuration with optional per-session override. |
| **BR-BKG-03** | Default cancellation cutoff is 4 hours before the session and is configurable. |
| **BR-BKG-04** | Waitlist is enabled. When a seat opens the next eligible waitlisted member is promoted automatically according to queue order. |
| **BR-BKG-05** | Members may self-book. Admin and Front Desk may book on behalf of members. Trainers may assist only when the permission has been granted. |
| **BR-BKG-06** | No role may bypass membership eligibility, payment or capacity rules. |

## 13.3 Attendance

| ID | Rule |
|---|---|
| **BR-ATT-01** | Attendance is recorded per class session or event session, not per day. |
| **BR-ATT-02** | Supported capture methods are QR check-in, trainer marking and audited administrative correction. |
| **BR-ATT-03** | Every correction records actor, timestamp, previous value, new value and reason, and the entry is immutable. |
| **BR-ATT-04** | A member cannot have two attendance records for the same session. |
| **BR-ATT-05** | Trainers may mark attendance only for sessions assigned to them. |

## 13.4 Events

| ID | Rule |
|---|---|
| **BR-EVT-01** | Default categories are Workshop, Internal Studio Event, Public Event and Special Session; categories are configurable. |
| **BR-EVT-02** | Events may be free or paid and open to the public, members only, or prospects. |
| **BR-EVT-03** | Each event stores capacity, date and time, registration window, price or payment requirement and audience type. |
| **BR-EVT-04** | A paid registration is confirmed only after verified payment; seat limits are always enforced. |

## 13.5 Payments

| ID | Rule |
|---|---|
| **BR-PAY-01** | The payment gateway is Razorpay; test mode is used until production credentials are supplied by the business. |
| **BR-PAY-02** | Only a verified server-side confirmation marks a payment as paid; front-end success indications are never trusted. |
| **BR-PAY-03** | Payment confirmation and membership approval are separate states. |
| **BR-PAY-04** | Each payment records internal identifier, provider, provider reference, amount and currency (INR default), status, purpose and timestamps. |
| **BR-PAY-05** | Gateway callbacks are idempotent; a delayed callback leaves the payment unresolved rather than assumed. |
| **BR-PAY-06** | Refunds and manual adjustments require permission and are audited. |

## 13.6 CRM

| ID | Rule |
|---|---|
| **BR-CRM-01** | Advertisement and landing-page submissions create a Lead or Workshop Registrant, never a Member. |
| **BR-CRM-02** | A lead becomes a member only after a defined conversion event such as membership purchase; source, campaign and history are preserved. |
| **BR-CRM-03** | Duplicate leads (same phone or email) are flagged or merged. |
| **BR-CRM-04** | Lead follow-up messages are subject to consent and template rules. |

## 13.7 Notifications

| Message | Default timing (all configurable) |
|---|---|
| **Registration, workshop and event registration confirmation** | Immediately |
| **Payment received** | Immediately after confirmed payment |
| **Membership approved or activated** | Immediately |
| **Booking confirmation** | Immediately |
| **Class reminders** | 12 hours and 1 hour before |
| **Event reminders** | 24 hours, 6 hours and 1 hour before |
| **Workshop reminders** | 24 hours, 12 hours and 1 hour before |
| **Renewal reminders** | 7, 3 and 1 days before expiry; expiry notice on the day |
| **Inactive member nudges (Phase 2)** | After 3 days and 7 days of inactivity |

| ID | Rule |
|---|---|
| **BR-NTF-01** | The workflow engine orchestrates; the messaging provider delivers; delivery or failure status is stored for every message. |
| **BR-NTF-02** | Messages are sent only after the business transaction commits; a failed message never reverses a valid transaction. |
| **BR-NTF-03** | Each event and recipient produces at most one message (idempotency). |
| **BR-NTF-04** | Until a production messaging account exists, test or mock configuration is used and outbound records are stored. |
| **BR-NTF-05** | Only approved templates are used; marketing-type messages require opt-in; opt-outs are honoured. |

## 13.8 AI Features

| ID | Rule |
|---|---|
| **BR-AI-01** | AI acts only through governed capabilities; it never reads or writes the database directly. |
| **BR-AI-02** | AI capabilities inherit the signed-in user's permissions and tenant scope. |
| **BR-AI-03** | Transactional actions require explicit user confirmation before execution. |
| **BR-AI-04** | The Assistant never guesses missing mandatory business data; it asks for it. |
| **BR-AI-05** | The Assistant reports the actual result returned by the platform and never claims success it did not receive. |
| **BR-AI-06** | Every AI action is logged with initiator, capability, arguments, result and timestamp. |
| **BR-AI-07** | Conversation context is temporary and becomes business data only through a governed capability. |
| **BR-AI-08** | The Enquiry Assistant answers from approved knowledge only and escalates when unsure. |
| **BR-AI-09** | Recommendations respect real eligibility and availability; no medical or health advice is given. |
| **BR-AI-10** | Advertising publication and budget spend require explicit human approval. |
| **BR-AI-11** | AI features are gated by subscription tier and feature flag per tenant. |

## 13.9 Platform and General

| ID | Rule |
|---|---|
| **BR-GEN-01** | Upstream ERPNext code is never modified; product logic lives in the custom application. |
| **BR-GEN-02** | Every business record is tenant-scoped and cross-tenant access is forbidden. |
| **BR-GEN-03** | Critical history is archived rather than destructively deleted. |
| **BR-GEN-04** | Identifiers are generated by the system; clients never supply identifiers on creation. |
| **BR-GEN-05** | No studio-specific name, logo, plan, class, reminder time or commercial rule is hard-coded where configuration can serve. |

# 14. Success Metrics and KPIs

Targets marked *baseline* are measured during the pilot and then set with the business; numeric targets elsewhere are proposed initial targets to be validated.

## 14.1 Business KPIs

| KPI | Definition | Target |
|---|---|---|
| **Active tenants** | Studios actively using the platform | Per commercial plan |
| **Recurring revenue** | Monthly subscription revenue plus add-ons | Per commercial plan |
| **Onboarding by configuration** | New studios onboarded without code changes | 90% or more |
| **Pilot acceptance** | Pilot studio sign-off on Phase 1 scope | Achieved |
| **Add-on attach rate** | Tenants using AI or WhatsApp add-ons | Baseline |

## 14.2 Operational KPIs

| KPI | Definition | Target |
|---|---|---|
| **Manual admin time** | Time spent on reminders, lookups and reconciliation | Measured reduction vs pilot baseline |
| **Time to register a member** | From start of registration to saved record | Under 1 minute |
| **Duplicate-record rate** | Duplicate members created per 100 registrations | Under 1 |
| **Exception resolution time** | Time to resolve flagged operational exceptions | Baseline |
| **Defects** | Sev-1 defects at release | Zero |

## 14.3 Membership KPIs

| KPI | Definition | Target |
|---|---|---|
| **Active, pending and expired members** | Counts by status | Reported daily |
| **Renewal rate** | Expiring memberships renewed within grace period | Baseline, then improve |
| **Reminder-to-renewal conversion** | Renewals within 7 days of a reminder | Baseline, then improve |
| **Approval turnaround** | Pending to active elapsed time | Baseline |

## 14.4 Attendance KPIs

| KPI | Definition | Target |
|---|---|---|
| **Class occupancy** | Bookings divided by capacity | Baseline, then improve |
| **Attendance vs bookings** | Attended sessions divided by booked | Baseline, then improve |
| **No-show rate** | Booked but not attended | Baseline, then reduce |
| **Attendance capture completeness** | Sessions with complete attendance records | 98% or more |

## 14.5 Revenue KPIs

| KPI | Definition | Target |
|---|---|---|
| **Revenue by plan, class and period** | Recognised member and event revenue | Reported monthly |
| **Payment reconciliation accuracy** | Payments matched to gateway records | 99% or more |
| **Payment exception rate** | Failed or mismatched payments | Baseline, then reduce |
| **Lead-to-member conversion** | Converted leads divided by leads | Baseline, then improve |

## 14.6 AI Adoption KPIs

| KPI | Definition | Target |
|---|---|---|
| **Assistant query success** | Queries answered correctly without escalation | 85% or more |
| **Assistant action accuracy** | Confirmed actions correct on first attempt | 95% or more |
| **False-success incidents** | Assistant claims success not returned by platform | Zero |
| **Assistant weekly active users** | Admins using the Assistant weekly | Baseline, then grow |
| **Enquiry deflection** | Enquiries resolved without a human | Baseline |
| **Recommendation acceptance** | Suggested classes subsequently booked | Baseline |

## 14.7 Automation KPIs

| KPI | Definition | Target |
|---|---|---|
| **Automation coverage** | Confirmations and reminders sent without staff action | 90% or more |
| **Delivery success** | Messages delivered with production account | 95% or more |
| **Failure visibility** | Failed workflow runs surfaced to admins | 100% |
| **Duplicate messages** | Duplicate sends per event | Zero |

# 15. Risks and Assumptions

## 15.1 Risks and Mitigation Plans

| ID | Risk | Impact | Mitigation |
|---|---|---|---|
| **R-01** | The 5-day scope is too large for the team. | High | Strict P1 cut; vertical slices; documented cut order in which AI is dropped first. |
| **R-02** | ERP customisation and learning curve slow delivery. | High | Isolated custom application; focused, time-boxed spikes; frozen Phase 1 data structures. |
| **R-03** | Extra integration hop between the API layer and the ERP adds complexity. | Medium | Thin adapter; interface contracts defined first; automated integration tests. |
| **R-04** | Business rules are incomplete. | Medium | Capture core rules, document assumptions, favour configuration, validate through demonstrations. |
| **R-05** | AI performs an incorrect or unintended action. | High | Confirmation, permission inheritance, backend validation, curated tools, audit. |
| **R-06** | AI produces inaccurate answers. | Medium | Answers built only from returned data; approved knowledge only; refuse rather than guess. |
| **R-07** | Production messaging account is unavailable. | Medium | Test and mock mode; stored delivery status; credential swap without code change. |
| **R-08** | Payment confirmation mismatches. | High | Verified callbacks, idempotency, reconciliation reports. |
| **R-09** | Cross-tenant data exposure. | High | Tenant-scoped access everywhere, isolation tests, least-privilege identities. |
| **R-10** | Dependence on and cost of a commercial AI provider. | Medium | Model-agnostic layer; evaluate open-source or local model. |
| **R-11** | Scope creep into enterprise features or vision-based coaching. | Medium | Phase and Future classification; documented change control. |
| **R-12** | Poor data quality limits AI value. | Medium | Validation at every capability; start with rules and retrieval. |
| **R-13** | Small team with uneven experience. | Medium | Feature-by-feature collaboration; clear ownership; common Definition of Done. |
| **R-14** | Hosting platform constraints on custom ERP applications. | Low-Medium | Confirm custom application support early. |
| **R-15** | Data protection non-compliance. | High | Consent, minimisation, retention rules, legal review. |

## 15.2 Constraints

- Upstream ERPNext code cannot be modified; product logic lives in a separate custom application.
- AI and automation must never access the database directly.
- Phase 1 is a five-day delivery by a small team.
- Production payment and messaging credentials are not yet available; test and mock configurations are used.
- Movement and posture intelligence is excluded while concept, intellectual-property and validation questions remain unresolved.
- Local development environments are lightweight; use managed ERP hosting rather than duplicating heavy runtimes.
- Secrets, dependency folders, virtual environments and backups are never committed to source control.

## 15.3 Assumptions

| ID | Assumption |
|---|---|
| **A-01** | ERPNext on Frappe is the final ERP choice for the beta; the alternative under earlier evaluation is retired. |
| **A-02** | A single API layer mediates between the front ends and the ERP (see Appendix A, conflict C-04); the mentor confirms this. |
| **A-03** | Front ends, API layer and workflow engine run on the team's cloud environment; the ERP runs on managed ERPNext hosting. |
| **A-04** | One pilot studio is used for the beta on the multi-tenant model. |
| **A-05** | Test credentials for the payment gateway and messaging are available; production credentials follow later. |
| **A-06** | A commercial foundation model is used in Phase 1; no model training or large dataset is required. |
| **A-07** | India-first launch: INR and English only. |
| **A-08** | Default rules (capacity 30, 4-hour cancellation cutoff, reminder timings) are development defaults pending client validation. |
| **A-09** | The Assistant's initial query set is agreed jointly by the business owner and the AI Assistant lead. |
| **A-10** | Approval authority is the mentor together with the business or client stakeholder. |

# 16. Roadmap

The roadmap separates the committed 5-Day Delivery from later phases. Feature flow for every item: Business Requirement, Full-Stack Capability, Interface for AI and automation, AI capability, Automation, End-to-End Validation. AI is never bolted on at the end.

## 16.1 Phase 1 - 5-Day Delivery (Current MVP)

**Goal:** a demonstrable, multi-tenant-ready operational core on ERPNext with WhatsApp automation (test mode), payments (test mode) and a narrow, safe AI slice.

| Day | Platform deliverables | AI deliverables |
|---|---|---|
| **Day 1 - Foundation** | Frozen scope and interface contracts for members, memberships, classes, bookings, attendance, leads and payments; ERP environment with custom application; tenant, studio and branch structure; authentication and permission framework; audit log; repository and continuous integration. | Tool definitions for the Assistant; guardrail and prompt specification; approved FAQ source list. |
| **Day 2 - Members, Memberships, Configuration** | Members with duplicate control; membership plans, assignment, approval and expiry; Configuration Center basics (plans, class types, capacity, cancellation window, branding). | Assistant skeleton with model connection, tool-calling loop and permission pass-through. |
| **Day 3 - Classes, Bookings, Attendance** | Class types and sessions; booking with eligibility, capacity, cancellation and waitlist; QR and trainer attendance with audited correction; rosters. | Read tools wired: memberships expiring this week; classes with low attendance. |
| **Day 4 - Payments, CRM, Automation, Portals** | Payment checkout (test) with verified confirmation; payment records; lead capture and conversion; workflows for booking confirmation, registration confirmation and renewal reminder with mock delivery logging; dashboard KPIs; member portal (dashboard, membership, bookings, profile). | Create Member through chat with missing-field handling, confirmation and true result reporting; AI audit logging. |
| **Day 5 - Events, Hardening, Demo** | Events with registration and seat limits; end-to-end regression on the canonical flow; permission and tenant-isolation tests; defect fixing; demo data; deployment. | Enquiry, Feedback, Recommendation and Booking-suggestion assistants (narrow scope); AI test cases and recorded results. |

### Must be completed in the 5-Day Delivery

- Tenant, studio and branch structure, six-role access control and audit logging.
- Members, Memberships (manual renewal), Classes, Bookings, Attendance, Events (basic), Payments (test mode), CRM (capture to conversion).
- Configuration Center basics and tenant branding.
- Core dashboard KPIs and core reports.
- Workflow engine with WhatsApp mock delivery for the core confirmations and reminders.
- B2C portal: dashboard, membership, bookings, attendance, events, payments and profile.
- Admin AI Assistant: two read queries and Create Member with confirmation.
- End-to-end test of the canonical flow: Member, Membership, Class, Booking, Attendance, Payment, Notification, Dashboard.

### MVP AI capabilities (Phase 1)

- AI Assistant (Admin): read queries and Create Member.
- AI Enquiry Assistant (approved knowledge and lead capture).
- AI Feedback Assistant (classify and summarise).
- AI Recommendation Assistant (rules-constrained).
- AI Booking Assistant (guided suggestions only; in-chat booking is Phase 2).

### Cut order if time slips (drop from the top)

| Order | Item dropped |
|---|---|
| **1** | Booking suggestions and Recommendation Assistant |
| **2** | Feedback Assistant |
| **3** | Enquiry Assistant |
| **4** | Paid-event payment |
| **5** | Create Member action (read queries retained) |
| **6** | Read queries |
| **Never** | Core operational modules, permission framework, tenant isolation, audit logging |

### Deferred from Phase 1

- Freeze, credits, upgrade and downgrade; refunds, invoices and reconciliation; event waitlist; inactive-member nudges; email; self-service tenant billing; in-app notification centre; conversational booking; custom reports and exports; MFA enforcement (mandatory before production); production messaging and payment credentials.

### Phase 1 Definition of Done

- Each module has defined requirements, validation, permissions, tenant isolation, audit logging where required, loading, empty, error and permission-denied states, automated tests, and no console, type or lint errors.
- The Assistant answers agreed queries with figures identical to the dashboard and completes Create Member with confirmation, correct failure handling and audit.
- Test cases and results are recorded; secrets are not in source control; the demo runs from the deployed environment.

## 16.2 Phase 2 (indicative four to eight weeks after Phase 1)

- **Platform:** freeze, credits and upgrades; refunds, invoices and tax; reconciliation; event waitlist; blackout dates and substitution; trial flow and attribution reports; drill-down reports and exports; in-app notifications and communication preferences; email; inactive-member nudges; self-service onboarding and SaaS billing; MFA; production messaging and payment; security and performance review; production deployment.
- **AI:** conversational Booking Assistant; personalised recommendations; more Assistant queries and actions with an approval queue; admin insights; CRM Agent and Operations Agent (read and draft); basic session summaries; feedback trend view; evaluation of open-source or local model.
- **Enterprise readiness:** location-aware roles, schedules and reporting foundation.

## 16.3 Phase 3

- Marketing Agent, Advertising Agent (human approval for publication and spend), Growth Optimization Agent and Studio Insights Agent.
- Campaign generation and performance analysis; pricing and schedule recommendations; cross-agent workflows.
- Retention and demand forecasting once sufficient historical data exists.

## 16.4 Future Product Evolution

- Multi-branch and franchise management, inter-branch roaming, white-label branding and domains, hardware and turnstile integration, trainer commission and payroll.
- Native mobile applications, referral programmes, marketplace, community, gamification, regional languages and additional currencies.
- Session Summary Agent and richer member progress experiences.
- Research track (not committed): movement and posture intelligence for yoga and similar disciplines, only after concept, intellectual-property and validation questions are resolved and dedicated resourcing is approved.

## 16.5 Agents by Phase

| Agent or assistant | Phase |
|---|---|
| **Admin AI Assistant (queries and Create Member)** | Phase 1 (MVP) |
| **Enquiry, Booking, Recommendation and Feedback assistants** | Phase 1 (narrow); expanded in Phase 2 |
| **CRM Agent, Operations Agent** | Phase 2 |
| **Marketing Agent, Advertising Agent, Growth Optimization Agent, Studio Insights Agent** | Phase 3 |
| **Session Summary Agent** | Future |

# Appendix A - Conflict Resolution Register

Inputs to this document were produced over several iterations and disagreed in places. The table records each conflict, the recommended final direction and the justification. Precedence rule applied: team-agreed final decisions override earlier documents; the most recent discussion overrides original lifecycle plans.

| ID | Topic | Conflict | Final direction and justification |
|---|---|---|---|
| **C-01** | ERP platform | Early plans specified one ERP; later material said it was not frozen and an alternative was under evaluation; final team decisions select ERPNext on Frappe. | **ERPNext on Frappe.** Team-agreed decision; better customisation flexibility, API simplicity and development speed for the timeline. Evaluation is closed. |
| **C-02** | Automation and messaging | Early plans used a third-party automation intermediary; later material replaced it. | **n8n plus WhatsApp Cloud API directly.** Removes an intermediary, gives direct control of templates, delivery status and compliance. |
| **C-03** | Front end | Early plans used a content-management front end for the member portal; later plans use Next.js. | **Next.js with TypeScript** for both portals; content-management system only optionally for marketing pages. |
| **C-04** | API layer | The final decision record lists the front end and the ERP custom application only; the technical and rebuild material introduces a dedicated API layer between them. | **Single API layer (NestJS) between Next.js and the ERP.** Provides one place for authentication, tenancy, permissions, validation, audit and AI orchestration, and prevents browsers calling the ERP directly. Flagged for mentor confirmation. |
| **C-05** | Timeline | Earlier plan: 14-week MVP. Later decision: one-week first sprint. This document: 5-day delivery including AI. | **Phase 1 is a 5-day delivery** with a reduced core and a narrow AI slice; remaining scope moves to Phase 2, with an explicit cut order. |
| **C-06** | AI timing | Final decisions place AI in Phase 2; the latest discussion wants an Assistant demonstration in the beta. | **Include a constrained AI slice in Phase 1**, behind a feature flag and built only on interfaces that exist that week. AI is dropped first if the schedule slips. |
| **C-07** | Roles | Earlier: four roles. Later: six roles including Front Desk and Visitor. | **Six roles:** Owner, Admin, Front Desk, Trainer, Member, Visitor (plus an operator-side platform role). |
| **C-08** | Membership plans | Fixed Basic and Premium plans versus configurable plans. | **Two layers:** platform tiers Basic, Premium, Ultra for studios; studio-defined member plans, fully configurable. |
| **C-09** | Movement and posture coaching | Some material includes it in scope; others defer it; the latest direction removes it from committed scope. | **Excluded from all committed phases**; research track only. Highest complexity, does not address the core operational pain point, and concept and IP questions are unresolved. |
| **C-10** | Agent phasing | Some material places enquiry and summary agents in the first phase and CRM and operations agents next; final decisions place all agents after the operational sprint. | Enquiry assistant is included narrowly in Phase 1; Session Summary Agent is Future; CRM and Operations agents are Phase 2; growth agents Phase 3. |
| **C-11** | Database | Application data in PostgreSQL versus ERP database choice. | **PostgreSQL for application data; ERPNext keeps its default database.** Avoids depending on less mature ERP support for the alternative database. |
| **C-12** | Capacity | '30 if required' versus fully configurable. | **Default 30, configurable per class or session**; never hard-coded. |
| **C-13** | Payments | Gateway to be decided versus selected. | **Razorpay** (test mode in Phase 1). Only verified server-side confirmation marks payment paid. |
| **C-14** | Enterprise features | Multi-branch, franchise, marketplace, roaming and white-label appear alongside earlier deferral of marketplace and native apps. | **Data model is multi-tenant and branch-ready from day one; enterprise features are Future.** |
| **C-15** | Hosting | Different hosting descriptions across material. | **Managed Frappe hosting for the ERP; the team's cloud environment for front ends, API layer, workflow engine and later AI services.** |

# Appendix B - Glossary

| Term | Meaning |
|---|---|
| **Tenant** | A studio business subscribing to the platform; its data, configuration and branding are isolated. |
| **B2B / B2C** | Studio management portal for staff / member portal for members and visitors. |

| Term | Meaning |
|---|---|
| **ERP** | Enterprise Resource Planning system; here ERPNext on the Frappe framework, the system of record. |
| **Governed capability** | A controlled, validated, permission-checked and audited operation exposed by the platform; used by the UI, AI and automation alike. |
| **Agent-ready** | Architecture in which AI agents can be added as consumers of governed capabilities without redesign. |
| **Workflow engine** | n8n; orchestrates event-driven, repeatable workflows; not a source of business truth. |
| **Lead / Registrant** | A prospect or workshop registrant who is not yet a member. |
| **Canonical flow** | Member, Membership, Class, Booking, Attendance, Payment, Notification, Dashboard. |
| **Configuration Center** | The module where tenants manage plans, class types, capacities, templates, timings and branding without code. |
| **Idempotent** | Repeating the same request or callback does not change the result beyond the first application. |

# Appendix C - Open Decisions for Mentor and Business

- Confirm the single API layer between the front ends and the ERP (C-04).
- Confirm the Phase 1 five-day cut and the AI slice, including the cut order (C-05, C-06).
- Validate default rules with the client: capacity 30, 4-hour cancellation cutoff, reminder timings.
- Identify the pilot studio and who supplies payment and messaging production credentials.
- Choose the beta AI provider and confirm data-privacy constraints for member data sent to it.
- Finalise the split of features and pricing across Basic, Premium and Ultra tiers.
- Decide whether non-member drop-in attendance is permitted.
- Complete legal review of data-protection consent and messaging opt-in.